

SIMATS ENGINEERING
ASSESSMENT TOOL 1 - High (INDUSTRY PROBLEM SOLVING
TASKS)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO4: Design and develop GUI-based user interfaces using Abstract Window Toolkit, Swing, and Applets by implementing event handling, layout managers, AWT controls, and menus. (BL3,BL5)

Assessment Tool Weightage: 10%

QUESTIONS: 5

Total Marks: 50

ANNEXURE A

INDUSTRY PROBLEM SOLVING TASKS

Code of Conduct:

I Challa Priyadarshini(192311118)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: CH Priyadarshini

ANNEXURE

INDUSTRY PROBLEM SOLVING TASKS

1. Online Examination System Interface (BTL: BL5 – Evaluate)

Design a Swing-based GUI application for conducting online exams, including features like question navigation, timer display, and result submission. Use components such as radio buttons, panels, and menus. Implement event handling for answering questions and submitting responses. Validate user actions to prevent multiple submissions. Evaluate layout managers for organizing question panels and controls. Analyze usability and responsiveness under multiple users and propose enhancements for reliability and scalability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution

		industry standards			
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Program:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.util.ArrayList;

public class OnlineExam extends JFrame implements ActionListener {
    JLabel titleLabel;
    JLabel timerLabel;
    JLabel questionNumberLabel;
    JLabel statusLabel;
    JTextArea questionArea;
    JRadioButton optionA;
    JRadioButton optionB;
    JRadioButton optionC;
    JRadioButton optionD;
    ButtonGroup optionGroup;
    JButton previousButton;
    JButton nextButton;
    JButton submitButton;
    JMenuBar menuBar;
    JMenu fileMenu;
    JMenu examMenu;
    JMenu helpMenu;
    JMenuItem restartItem;
    JMenuItem submitMenuItem;
    JMenuItem exitItem;
    JMenuItem instructionsItem;
    JPanel questionPanel;
    JPanel optionPanel;
    JPanel navigationPanel;
    JPanel bottomPanel;
    Timer timer;
    String[] questions = {
        "Which language is primarily used for Android development?",
        "Which keyword is used to inherit a class in Java?",
        "Which collection does not allow duplicate elements?",
        "Which component is used to create a window in Swing?",
        "Which method is used to start a Java thread?"
    };
    String[][] options = {
        {
            "Java",
            "HTML",
            "SQL",
```

```

        "CSS"
    },
    {
        "implements",
        "extends",
        "inherits",
        "super"
    },
    {
        "ArrayList",
        "LinkedList",
        "HashSet",
        "Vector"
    },
    {
        "JFrame",
        "JButton",
        "JLabel",
        "JPanel"
    },
    {
        "run()",
        "start()",
        "begin()",
        "execute()"
    }
};
int[] correctAnswers = {
    0,
    1,
    2,
    0,
    1
};
int[] userAnswers = {
    -1,
    -1,
    -1,
    -1,
    -1
};
int currentQuestion = 0;
int timeRemaining = 300;

```

```

boolean submitted = false;
Color creamBackground =
    new Color(250, 244, 230);
Color darkGreen =
    new Color(35, 90, 60)
Color lightGreen =
    new Color(220, 240, 225);
Color maroon =
    new Color(125, 45, 55);
Color lightMaroon =
    new Color(245, 225, 225);
Color orange =
    new Color(220, 125, 45);
Color darkText =
    new Color(45, 45, 45);
public OnlineExam() {
    setTitle("Online Examination System");
    setSize(900, 650);
    setLocationRelativeTo(null);
    setDefaultCloseOperation(
        JFrame.EXIT_ON_CLOSE
    );
    createMenu();
    createInterface();
    displayQuestion();
    startTimer();
    setVisible(true);
}
private void createMenu() {
    menuBar = new JMenuBar();
    fileMenu = new JMenu("File");
    examMenu = new JMenu("Exam");
    helpMenu = new JMenu("Help");
    restartItem =
        new JMenuItem("Restart Exam");
    submitMenuItem =
        new JMenuItem("Submit Exam");
    exitItem =
        new JMenuItem("Exit");
    instructionsItem =
        new JMenuItem("Instructions");
    restartItem.addActionListener(this);
    submitMenuItem.addActionListener(this);
    exitItem.addActionListener(this);
}

```

```

instructionsItem.addActionListener(this);
fileMenu.add(restartItem);
fileMenu.addSeparator();
fileMenu.add(exitItem);
examMenu.add(submitMenuItem);
helpMenu.add(instructionsItem);
menuBar.add(fileMenu);
menuBar.add(examMenu);
menuBar.add(helpMenu);
setJMenuBar(menuBar);
}

```

```

private void createInterface() {
    JPanel mainPanel =
        new JPanel(new BorderLayout(15, 15));
    mainPanel.setBackground(
        creamBackground
    );
    mainPanel.setBorder(
        BorderFactory.createEmptyBorder(
            15, 15, 15, 15
        )
    );
    JPanel headerPanel =
        new JPanel(new BorderLayout());

    headerPanel.setBackground(
        darkGreen
    );
    titleLabel =
        new JLabel(
            "ONLINE EXAMINATION SYSTEM",
            JLabel.CENTER
        );
    titleLabel.setFont(
        new Font(
            "Georgia",
            Font.BOLD,
            28
        )
    );

    titleLabel.setForeground(

```

```
        Color.WHITE
    );
    timerLabel =
        new JLabel(
            "Time Left: 05:00",
            JLabel.CENTER
        );
    timerLabel.setFont(
        new Font(
            "Arial",
            Font.BOLD,
            18
        )
    );
    timerLabel.setForeground(
        Color.YELLOW
    );
    headerPanel.add(
        titleLabel,
        BorderLayout.CENTER
    );
    headerPanel.add(
        timerLabel,
        BorderLayout.EAST
    );
    mainPanel.add(
        headerPanel,
        BorderLayout.NORTH
    );
    questionPanel =
        new JPanel(
            new BorderLayout(10, 10)
        );
    questionPanel.setBackground(
        Color.WHITE
    );
    questionPanel.setBorder(
        BorderFactory.createCompoundBorder(
            BorderFactory.createLineBorder(
                darkGreen,
                2
            ),
            BorderFactory.createEmptyBorder(
                15, 15, 15, 15
            )
        )
    );
```



```

        )
    )
);
questionNumberLabel =
    new JLabel();
questionNumberLabel.setFont(
    new Font(
        "Arial",
        Font.BOLD,
        19
    )
);
questionNumberLabel.setForeground(
    maroon
);
questionArea =
    new JTextArea();
questionArea.setFont(
    new Font(
        "Arial",
        Font.BOLD,
        20
    )
);

questionArea.setForeground(
    darkText
);
questionArea.setBackground(
    new Color(255, 252, 245)
);
questionArea.setLineWrap(true);
questionArea.setWrapStyleWord(true);
questionArea.setEditable(false);
questionArea.setBorder(
    BorderFactory.createEmptyBorder(
        10, 10, 10, 10
    )
);
questionPanel.add(
    questionNumberLabel,
    BorderLayout.NORTH
);
questionPanel.add(

```

```
        questionArea,
        BorderLayout.CENTER
    );
    optionPanel =
        new JPanel(
            new GridLayout(
                4,
                1,
                10,
                10
            )
        );
    optionPanel.setBackground(
        lightGreen
    );
    optionA =
        new JRadioButton();
    optionB =
        new JRadioButton();
    optionC =
        new JRadioButton();
    optionD =
        new JRadioButton();
    optionA.setFont(
        new Font(
            "Arial",
            Font.PLAIN,
            17
        )
    );
    optionB.setFont(
        new Font(
            "Arial",
            Font.PLAIN,
            17
        )
    );
    optionC.setFont(
        new Font(
            "Arial",
            Font.PLAIN,
            17
        )
    );
    );
```

```

optionD.setFont(
    new Font(
        "Arial",
        Font.PLAIN,
        17
    )
);
optionA.setBackground(
    lightGreen
);
optionB.setBackground(
    lightGreen
);
optionC.setBackground(
    lightGreen
);
optionD.setBackground(
    lightGreen
);
optionA.setForeground(darkText);
optionB.setForeground(darkText);
optionC.setForeground(darkText);
optionD.setForeground(darkText);
optionGroup =
    new ButtonGroup();
optionGroup.add(optionA);
optionGroup.add(optionB);
optionGroup.add(optionC);
optionGroup.add(optionD);
optionPanel.add(optionA);
optionPanel.add(optionB);
optionPanel.add(optionC);
optionPanel.add(optionD);
JPanel centerPanel =
    new JPanel(
        new BorderLayout(10, 10)
    );

centerPanel.setBackground(
    creamBackground
);
centerPanel.add(
    questionPanel,
    BorderLayout.NORTH

```

```

);
centerPanel.add(
    optionPanel,
    BorderLayout.CENTER
);
mainPanel.add(
    centerPanel,
    BorderLayout.CENTER
);
navigationPanel =
    new JPanel(
        new FlowLayout(
            FlowLayout.CENTER,
            15,
            10
        )
    );
navigationPanel.setBackground(
    creamBackground
);
previousButton =
    new JButton("← Previous");
nextButton =
    new JButton("Next →");
submitButton =
    new JButton("SUBMIT EXAM");
styleButton(
    previousButton,
    darkGreen
);
styleButton(
    nextButton,
    orange
);
styleButton(
    submitButton,
    maroon
);
previousButton.addActionListener(this);
nextButton.addActionListener(this);
submitButton.addActionListener(this);
navigationPanel.add(
    previousButton
);

```

```

navigationPanel.add(
    nextButton
);
navigationPanel.add(
    submitButton
);
statusLabel =
    new JLabel(
        "Question 1 of 5",
        JLabel.CENTER
    );
statusLabel.setFont(
    new Font(
        "Arial",
        Font.BOLD,
        16
    )
);
statusLabel.setForeground(
    darkGreen
);
bottomPanel =
    new JPanel(
        new BorderLayout()
    );
bottomPanel.setBackground(
    creamBackground
);
bottomPanel.add(
    navigationPanel,
    BorderLayout.CENTER
);
bottomPanel.add(
    statusLabel,
    BorderLayout.SOUTH
);
mainPanel.add(
    bottomPanel,
    BorderLayout.SOUTH
);
add(mainPanel);
}
private void styleButton(
    JButton button,

```

```

        Color color
    ) {
        button.setFont(
            new Font(
                "Arial",
                Font.BOLD,
                16
            )
        );
        button.setBackground(color);
        button.setForeground(
            Color.WHITE
        );
        button.setFocusPainted(false);
        button.setBorder(
            BorderFactory.createEmptyBorder(
                10, 20, 10, 20
            )
        );
    }

    private void displayQuestion() {
        questionNumberLabel.setText(
            "Question "
            + (currentQuestion + 1)
            + " of "
            + questions.length
        );
        questionArea.setText(
            questions[currentQuestion]
        );
        optionA.setText(
            "A. " + options[currentQuestion][0]
        );
        optionB.setText(
            "B. " + options[currentQuestion][1]
        );
        optionC.setText(
            "C. " + options[currentQuestion][2]
        );
        optionD.setText(
            "D. " + options[currentQuestion][3]
        );
        optionGroup.clearSelection();
        if (userAnswers[currentQuestion] == 0) {

```

```

        optionA.setSelected(true);
    }
    else if (userAnswers[currentQuestion] == 1) {
        optionB.setSelected(true);
    }
    else if (userAnswers[currentQuestion] == 2) {
        optionC.setSelected(true);
    }
    else if (userAnswers[currentQuestion] == 3) {
        optionD.setSelected(true);
    }
    statusLabel.setText(
        "Question "
        + (currentQuestion + 1)
        + " of "
        + questions.length
    );
    previousButton.setEnabled(
        currentQuestion > 0
    );

    nextButton.setEnabled(
        currentQuestion <
        questions.length - 1
    );
}
private void saveAnswer() {
    if (optionA.isSelected()) {
        userAnswers[currentQuestion] = 0;
    }
    else if (optionB.isSelected()) {
        userAnswers[currentQuestion] = 1;
    }
    else if (optionC.isSelected()) {
        userAnswers[currentQuestion] = 2;
    }
    else if (optionD.isSelected()) {
        userAnswers[currentQuestion] = 3;
    }
}
private void startTimer() {
    timer =
        new Timer(
            1000,

```

```

new ActionListener() {
    public void actionPerformed(
        ActionEvent e
    ) {

        if (submitted) {
            timer.stop();
            return;
        }
        timeRemaining--;
        int minutes =
            timeRemaining / 60;
        int seconds =
            timeRemaining % 60;
        timerLabel.setText(
            String.format(
                "Time Left: %02d:%02d",
                minutes,
                seconds
            )
        );
        if (timeRemaining <= 30) {
            timerLabel.setForeground(
                Color.RED
            );
        }
        if (timeRemaining <= 0) {
            timer.stop();
            JOptionPane.showMessageDialog(
                OnlineExam.this,
                "Time is over. Your exam will be submitted automatically.",
                "Time Over",
                JOptionPane.WARNING_MESSAGE
            );
            submitExam();
        }
    }
}

);
timer.start();
}

private void submitExam() {
    if (submitted) {
        JOptionPane.showMessageDialog(

```



```

        this,
        "Your exam has already been submitted.",
        "Submission Blocked",
        JOptionPane.WARNING_MESSAGE
    );
    return;
}
saveAnswer();
int unanswered = 0;
for (int answer : userAnswers) {
    if (answer == -1) {
        unanswered++;
    }
}
if (unanswered > 0) {
    int choice =
        JOptionPane.showConfirmDialog(
            this,
            unanswered
            + " question(s) are unanswered.\n"
            + "Do you still want to submit?",
            "Confirm Submission",
            JOptionPane.YES_NO_OPTION
        );
    if (choice != JOptionPane.YES_OPTION) {
        return;
    }
}
int score = 0;
for (int i = 0;
     i < questions.length;
     i++) {
    if (userAnswers[i] ==
        correctAnswers[i]) {
        score++;
    }
}
submitted = true;
if (timer != null) {
    timer.stop();
}
previousButton.setEnabled(false);
nextButton.setEnabled(false);
submitButton.setEnabled(false);

```

```

optionA.setEnabled(false);
optionB.setEnabled(false);
optionC.setEnabled(false);
optionD.setEnabled(false);
String result =
    "EXAM SUBMITTED SUCCESSFULLY\n\n"
    + "Total Questions : "
    + questions.length
    + "\n"
    + "Correct Answers : "
    + score
    + "\n"
    + "Wrong / Unanswered : "
    + (questions.length - score)
    + "\n"
    + "Score : "
    + score
    + " / "
    + questions.length;
JOptionPane.showMessageDialog(
    this,
    result,
    "Exam Result",
    JOptionPane.INFORMATION_MESSAGE
);
statusLabel.setText(
    "EXAM SUBMITTED - Score: "
    + score
    + "/"
    + questions.length
);
}
private void restartExam() {
    if (submitted) {
        int choice =
            JOptionPane.showConfirmDialog(
                this,
                "Start a new exam?",
                "Restart Exam",
                JOptionPane.YES_NO_OPTION
            );
        if (choice != JOptionPane.YES_OPTION) {
            return;
        }
    }
}

```

```

    }
    for (int i = 0;
        i < userAnswers.length;
        i++) {
        userAnswers[i] = -1;
    }
    currentQuestion = 0;
    timeRemaining = 300;
    submitted = false;
    timerLabel.setForeground(
        Color.YELLOW
    );
    submitButton.setEnabled(true);
    displayQuestion();
    startTimer();
}
public void actionPerformed(
    ActionEvent e
) {
    Object source =
        e.getSource();
    if (source == previousButton) {
        saveAnswer();
        if (currentQuestion > 0) {

            currentQuestion--;
            displayQuestion();
        }
    }
    else if (source == nextButton) {
        saveAnswer();
        if (currentQuestion <
            questions.length - 1) {
            currentQuestion++;
            displayQuestion();
        }
    }
    else if (source == submitButton) {
        submitExam();
    }
    else if (source == submitMenuItem) {
        submitExam();
    }
    else if (source == restartItem) {

```

```

        restartExam();
    }
    else if (source == exitItem) {
        System.exit(0);
    }
    else if (source == instructionsItem) {
        JOptionPane.showMessageDialog(
            this,
            "1. Select an answer.\n"
            + "2. Use Next and Previous to navigate.\n"
            + "3. You can change an answer before submission.\n"
            + "4. Submit the exam before the timer ends.\n"
            + "5. Multiple submissions are prevented.",
            "Exam Instructions",
            JOptionPane.INFORMATION_MESSAGE
        );
    }
}

public static void main(String[] args) {
    SwingUtilities.invokeLater(
        new Runnable() {
            public void run() {
                new OnlineExam();
            }
        }
    );
}
}

```

OUTPUT:

Online Examination System

File Exam Help

ONLINE EXAMINATION SYSTEM

Time Left: 04:50

Question 1 of 5

Which language is primarily used for Android development?

☐ A. Java

☒ B. HTML

☐ C. SQL

☐ D. CSS

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SUBMIT EXAM

Question 1 of 5

Online Examination System

File Exam Help

ONLINE EXAMINATION SYSTEM

Time Left: 04:36

Question 2 of 5

Which keyword is used to inherit a class in Java?

☐ A. implements

☐ B. extends

☒ C. inherits

☐ D. super

← Previous

Next →

SUBMIT EXAM

Question 2 of 5

Online Examination System

File Exam Help

ONLINE EXAMINATION SYSTEM

Time Left: 04:16

Question 3 of 5

Which collection does not allow duplicate elements?

- ☐ A. ArrayList
- ☐ B. LinkedList
- ☒ C. HashSet
- ☐ D. Vector

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Question 3 of 5

Online Examination System

File Exam Help

ONLINE EXAMINATION SYSTEM

Time Left: 04:04

Question 4 of 5

Which component is used to create a window in Swing?

- ☒ A. JFrame
- ☐ B. JButton
- ☐ C. JLabel
- ☐ D. JPanel

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Question 4 of 5

Online Examination System

File Exam Help

ONLINE EXAMINATION SYSTEMTime Left: 03:55

Question 5 of 5

Which method is used to start a Java thread?

☐ A. run()

☒ B. start()

☐ C. begin()

☐ D. execute()

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SUBMIT EXAM

Question 5 of 5

Online Examination System

File Exam Help

ONLINE EXAMINATION SYSTEMTime Left: 04:35

Question 5 of 5

Which method is used to start a Java thread?

☐ A. run()

☒ B. start()

☐ C. begin()

☐ D. execute()

Exam Result

i

EXAM SUBMITTED SUCCESSFULLY

Total Questions : 5

Correct Answers : 4

Wrong / Unanswered : 1

Score : 4 / 5

OK

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SUBMIT EXAM

Question 5 of 5

Online Examination System

File Exam Help

ONLINE EXAMINATION SYSTEMTime Left: 03:49

Question 5 of 5

Which method is used to start a Java thread?

☐ A. run()

☒ B. start()

☐ C. begin()

☐ D. execute()

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SUBMIT EXAM

EXAM SUBMITTED - Score: 3/5

Online Examination System

File Exam Help

ONLINE EXAMINATION SYSTEMTime Left: 04:24

Question 5 of 5

Which method is used to start a Java thread?

☐ A. run()

☒ B. start()

☐ C. begin()

☐ D. execute()

Restart Exam

?

Start a new exam?

Yes

No

← Previous

Next →

SUBMIT EXAM

EXAM SUBMITTED - Score: 4/5

